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Chapter 6

Remote Communications



Remote Communications For Service Management

You can use the remote communications link to pass job and message data to field engineers, and receive details of work done back from them.

The information entered during the call logging and technical reporting procedures is stored in outgoing and incoming monitor files, subject to strict validation and error reporting routines. Data is supplied to a communications gateway which is flexible, to allow the link to be directly from a mid-range server, or from a personal computer.

A batch sleeper job extracts data to be transmitted to active outstations, and carries out all necessary file updates and status changes.

Engineers logging on to the system receive call details and messages; reports of work done are passed back to the centre. Engineers can accept, reject and otherwise change the status of jobs to reflect their current progress. They can report work done which incurs charges for labour, travel time and distance, parts used and returned, plus miscellaneous costs and charges.

They can also log and report on additional jobs. New and changed equipment details can update a site's installed equipment records. In addition, any related messages or text can be communicated.

Enquiries and reports warn you of any validation errors in incoming or outgoing information. Maintenance routines are available to correct the errors.

Engineers logging off, at day end, or for any other reason for absence, will be noted by the system. Depending on the type of log off, the jobs may be left for the same engineer to work on, or may be made available for reallocation to others.

Description

Call logging and technical reporting procedures are described in separate sections. Their cycles of operation remain very much as before. They interact with an extended range of job sub-status codes (parameter type ETAR on the codes/parameters file), remote communications error codes (parameter type ERRC), remote communications process indicators (parameter type PCDF) and transaction types (parameter type TTYP).

The operation of the remote communications link on the server, requires that the following steps are taken:

- Include detail in the branch record for the error message queue; the default otherwise is QSYSOPR.
- Set the Remote Communications Active field on in the System Parameters File Maintenance option (screen 4 of 5).
- Include any extra job sub-status codes in the Codes/Parameter File Maintenance option, parameter type ETAR.
- Make sure that the Service Management sub-system is active.

Implications

Hand-held terminal set-up instructions must be followed carefully: they are not detailed here.

You must make sure that any system-required codes are set up, that the incoming and outgoing files are updated correctly, and that the sub-system is active. Unpredictable results will follow if there are errors or omissions in these areas.

Normal operation of call logging and technical reporting can continue during remote communications operations. If remote communications is not active, standard operations, making use of the extended job sub-status codes, will continue.

Start Incoming Monitor



1/SSR



Select Start Incoming Monitor

Use this screen to start the Incoming Transaction Monitor (ITM).

```
SS862P          Z1 - C&T TEST COMPANY          USER GBJBABL    11/07/96
                                                    14:33:43

Start Incoming Monitor

You have requested to submit the Remote Communications Interface Incoming
Sleeper program. A previous request is still outstanding,
and may be active, or have terminated abnormally.
Press F16 to override and submit the sleeper.

Previous request outstanding:

Requested by.. GBJBABL
               QPADEV0005
At.....      14:33:25

F3=Exit  F16=Submit

01-02    [ ]    MW    KS    IM    II    KENB20A  PET178S2
```

Records are held on the incoming transactions file, sequenced by date and time of the transaction from the remote transmission. They may cover technical reporting, acknowledging receipt of job details, job acceptance, job referral, re-sequencing job queues, job status changes and engineer location.

Two fields on each record on the file are updated: they are the Processed Flag (parameter type PCDF in the Codes/Parameters File Maintenance option) and the Error Code (parameter type ERRC).

Successful transactions mean that file updates are made. An acknowledgement can be sent, if flagged as required, via the outgoing transactions file.

Partly successful transactions are flagged as partly processed, requires manual completion, with an error code to define the problem. A message is sent to the designated message queue, and the acknowledgement routine can be followed, as appropriate.

Rejected transactions are flagged as rejected, with an error code to define the problem. Validation continues until all errors are found.



Note: A number of error codes can be associated with each transaction.

A message is sent to the designated message queue, and the acknowledgement routine can be followed, as appropriate.

Held transactions are flagged. The condition arises when a transaction is valid, but there is an outstanding invalid transaction for the same engineer, job and

machine combination. Any acknowledgement will be sent, but will indicate that the record is being held.

Completed transactions are removed from the incoming transactions file whenever the audit report is printed, and are flagged to purge successful and deleted records (see options 5 and 7).



Select **F16=Submit** to activate the monitor. It runs continuously as a background batch job, processing each transaction as it arrives at the incoming transaction file. After processing all outstanding transactions, the program sleeps for the number of minutes specified in the parameter type WAIT in the Codes/Parameter File Maintenance option.

Start Outgoing Monitor



2/SSR



Select Start Outgoing Monitor

Use this screen to start the Outgoing Transaction Monitor (OTM).

```
SS860P          Z1 - UK Demonstration Company  USER GBJBALTW  21/05/98
                                                    10:11:36

Start Outgoing Monitor

You have requested to submit the Remote Communications Interface Outgoing
Sleeper program. A previous request is still outstanding,
and may be active, or have terminated abnormally.
Press F16 to override and submit the sleeper.

Previous request outstanding:

                Requested by.. ENDRQS
                QPADEV0012
                At..... 18:40:58

F3=Exit  F16=Submit

MA  a                                                    01/001
```

The program, a sleeper job, runs in the background, monitoring both the job and the outgoing transaction files for changes which require processing.

As soon as a job is flagged for transmission, the OTM writes details to the outgoing transaction file, but only if the engineer is logged on.

The OTM updates the job file to reflect the transmission status (parameter type TTPY in the Codes/Parameter File Maintenance option) of jobs which have already been transferred. Each transaction written, is given a sequential number, and is dated and timed.

Outgoing transactions include job details notification (new or changed), job transfers, job deallocated, job cancelled, acknowledge receipt of engineer transmission, and so on.



Select **F8=Submit** to run the program which controls the processing of all outgoing transactions.

End Incoming Monitor

1...
2...

3/SSR



Take the End Incoming Monitor option to display the screen to close down the Incoming Transaction Monitor (ITM).

```
SS862E                Z1 - UK Demonstration Company    USER GBJBALTW    21/05/98
                                                              10:15:39

End Incoming Monitor

You have requested to terminate the Remote Communications Interface Incoming
Sleeper program. To proceed and end the program,
press F8 now. If you do not wish to end the program
at this stage, press F3 to return to the menu.

F3=Exit  F8=Terminate

MA  a                                01/001
```



Note: A message is displayed, allowing you to confirm your choice.



Select **F8=Terminate** to close down the Incoming Transaction Monitor (ITM) in a controlled fashion, after the current cycle is completed.

End Outgoing Monitor

1...
2...

4/SSR



Take the End Outgoing Monitor option to display the screen to close down the Outgoing Transaction Monitor (OTM).

```
SS860E                Z1 - UK Demonstration Company    USER GBJBALTW    21/05/98
                                                              10:16:56

End Outgoing Monitor

You have requested to terminate the Remote Communications Interface Outgoing
Sleeper program. To proceed and end the program,
press F8 now. If you do not wish to end the program
at this stage, press F3 to return to the menu.

F3=Exit  F8=Terminate

MA a                                01/001
```



Note: A message is displayed, allowing you to confirm your choice.



Select **F8=Terminate** to close down the Outgoing Transaction Monitor (OTM) in a controlled fashion, after the current cycle is completed.

Print Audit Report

1...
2...

5/SSR



Select Print Audit Report.

SS864P	Z1 - C&T TEST COMPANY	USER GBJBAJKG	25/04/96 15:55:55
Remote Communication - Request Incoming Transaction Reports / Purge			
Print Full Audit or Exceptions only..... @ (0 = Exceptions Only) (1 = Exceptions & Audit)			
Purge Transactions Once Printed..... @ (0 = No) (1 = Yes)			
F3=Exit F8=Submit			
07-46	SR	MM	KS IM II KENB20A PED129S1

Fields:

Full Audit or Exceptions

Enter one of the following:

0 - Print only transactions which are partly processed, rejected, held or deleted.

1 - Print not only those transactions which are partly processed, rejected, held or deleted, but also the fully processed records.



Note: Records awaiting processing are ignored.

Purge Transactions

Enter one of the following:

0 - No purging is to take place.

1 - Purge fully processed and deleted records once the report has been printed.



Select **F8=Submit** to send a batch job to create the report.

Purge Transaction Files



7/SSR



Select Purge Transaction Files

Use this option to submit a batch job to purge all fully processed and deleted incoming transaction records.